BATUHAN KARAMAN

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INTERESTS

LLM research & product Reasoning LLMs, RLVR/RLHF, Alignment, VLLMs, Agents, Responsible AI
Health AI Longitudinal data analysis, Clinical decision support, Medical Imaging

EDUCATION

Cornell University Ithaca, NY
Ph.D. in Machine Learning (ECE) 2020 - 2026
Advisors: Mert Sabuncu, Katerina Dodelzon (from Weill Cornell Medicine Radiology)

- My PhD research focused on developing time-series and computer vision methods to predict clinical outcomes from longitudinal multimodal biomedical data.

Middle East Technical University Ankara, Turkey
B.S. in Electrical Engineering 2015 - 2020

EXPERIENCE

Amazon Web Services (AWS) Health AI Sunnyvale, CA
Applied Scientist Aug 2026 - Present
Supervisors: Guoli Sun, Thayn Moore

- Working on responsible AI for multiple healthcare agents.

Amazon AGI Sunnyvale, CA
Applied Scientist Intern Jun 2025 - Sep 2025
Supervisors: Mohammad Ghavamzadeh, Arijit Biswas, Ruida Zhou

- Developed a novel REINFORCE-based RLVR algorithm that achieves state-of-the-art performance on various math reasoning benchmarks using Qwen- and Llama-family models.
- Analyzed the sources of instability in the REINFORCE Algorithm.

Microsoft AI Redmond, WA
Applied Scientist Intern Jun 2024 - Aug 2024
Supervisors: Xia Song, Alon Benhaim, Maggie Engler (previously at Inflection AI)

- Worked on the safety and usefulness balance of Llama- and Phi-family models through SFT and preference optimization.
- Developed a novel preference optimization method, achieving a 30% reduction in model overrefusals.

Spectral AI Dallas, TX (Remote)
Deep Learning Scientist Intern (Part-time) Aug 2023 - Nov 2023

- Designed a multimodal cross-attention-based model for diabetic foot ulcer healing prediction, combining multispectral imagery and clinical data. Achieved improved lesion localization.

Spectral AI Dallas, TX
Deep Learning Scientist Intern Jun 2023 - Aug 2023

- Enhanced a multimodal convolutional model for diabetic foot ulcer healing prediction, integrating multispectral imagery and clinical data. Improved classification accuracy by 8%.

PUBLICATIONS

1. **Karaman, B.K.**[†], Rawal, A., Shakiah S., Ghavamzadeh, M., Hong, M., Biswas, A., Zhou, R., "DISPO: Enhancing Training Efficiency and Stability in Reinforcement Learning for Large Language Model Mathematical Reasoning", AISTATS 2026. [Paper] [Code]
2. **Karaman, B.K.**, Feldman, P., Adap S., Dodelzon, K., Sabuncu, M.R., "LoMaR-v2: A Deep Survival Model for Breast Cancer Risk Prediction Using Longitudinal Mammography Data", Under Review, 2026.
3. **Karaman, B.K.**[‡], Zabir, I., Benhaim, A., Chaudhary V., Sabuncu, M.R., Song, X., "POROver: Improving Safety and Reducing Overrefusal in Large Language Models with Overgeneration and Preference Optimization", ICML 2025. [Paper]
4. **Karaman, B.K.***, Nguyen, M.*, Kim, H.*, Wang, A.Q.*, Liu, F.*, Sabuncu, M.R., "Knockout: A Simple Way to Handle Missing Inputs.", TMLR, 2025. [Paper] [Code]
5. **Karaman, B.K.**, Nguyen, M., Kim, H., Sabuncu, M.R., "A Deep Survival Model for Predicting Alzheimer's Diagnosis based on Multi-modal Longitudinal Data", IEEE BDMA, 2025. [Paper]
6. **Karaman, B.K.**, Sabuncu, M.R., "Assessing the Significance of Longitudinal Data in Alzheimer's Disease Forecasting", AIiH 2024 (Best Paper Award). [Paper]
7. **Karaman, B.K.**, Dodelzon, K., Akar, G.B., Sabuncu, M.R., "Longitudinal Mammogram Risk Prediction.", MICCAI 2024. [Paper] [Code]
8. Kim, H., **Karaman, B.K.**, Zhao, Q., Wang, A.Q., Sabuncu, M.R., "Learning-based Inference of Longitudinal Image Changes: Applications in Embryo Development, Wound Healing, and Aging Brain", PNAS, 2024. [Paper]
9. Wang A.Q., **Karaman B.K.**, Kim H., Rosenthal J., Saluja R., Young S.I., Sabuncu M.R., "A Framework for Interpretability in Machine Learning for Medical Imaging", IEEE Access, 2023. [Paper]
10. **Karaman B.K.**, Mormino E.C., Sabuncu M.R., "Machine learning based multi-modal prediction of future decline toward Alzheimer's disease: An empirical study", PLoS ONE, 2022. [Paper] [Code] (Highlighted at Cornell Chronicle on Nov 23rd, 2022. [Article])

INVITED TALKS & SYMPOSIUMS

1. Distinguished speaker at the 6th Global Conclave on Neurology and Neurological Disorders (NEURO Conclave 2025): "Assessing the Significance of Longitudinal Data in Alzheimer's Disease Forecasting".
2. Distinguished speaker at the 5th International Conference on Future of Preventive Medicine and Public Health (Future of PMPH 2025): "Longitudinal Mammogram Risk Prediction".
3. Machine Learning in Medicine Symposium (MLIM 2022): "Machine learning based multi-modal prediction of future decline toward Alzheimer's Disease".

HONORS & AWARDS

- Best Paper Award, International Conference on AI in Healthcare (AIiH), 2024.
- Irwin Jacobs Scholar Fellowship, Cornell University, 2020.
- METU High Honor Award, METU, 2020.

SERVICE

- **Reviewer:** ICML, NeurIPS, ICLR, AISTATS, Nature, PloS, MELBA, AIiH, Journal of Alzheimer's, etc.
- **Teaching:** Mentored multiple Cornell Master's and Undergraduate students on their graduation projects and theses.

[†] Work done during an internship at Amazon AGI.
[‡] Work done during an internship at Microsoft AI.
[*] Indicates equal contribution.